A square-zero gauge fixing for the heat-kernel track of $\mathcal{N}=4$ super Yang–Mills theory

formulations, exact checks, equivariant structure, migration

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Notation is that of the manuscript (Chapters 1, 2, 7, 8): $\mathscr{E}$ the BV fields $(c, A, \lambda; c^*, A^*, \lambda^*)$, Q of Proposition c1-Q, the pairing of Definition c1-pairing, K_0 of Definition c1-bracket, $v.$ of Definition c7-vdot, conformal Killing spinors $\epsilon = \xi + \rho(x)\eta$, Γ of [ESW, (28)]. Momentum space: $d \mapsto ip$, d^* on one-forms $\mapsto -i(p, -)$. The unit is $\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X \mathrm{ad}_Y)/(16\pi^2(X,Y))$. Every statement marked [check] is an exact computation of `checks.py`, recorded in `checks.json`.

1 The conditions on $\mathbb{R}^4$

Definition 1.1 (Admissible operators and the conditions)**.** An operator G on a BV complex $(\mathscr{F}, Q)$ over $\mathbb{R}^4$ is admissible if it is odd of degree -1, a sum of constant-coefficient differential operators of order ≤ 1 between slots, covariant under $\mathrm{Spin}(4,\mathbb{C}) \times \mathrm{Spin}(6,\mathbb{C})$, and homogeneous of Weyl weight 2 (a component from weight w to weight w' carries $w' - w + 2$ derivatives; no dimensionful constant). The conditions of [C07, Def 4.3.3] read on $\mathbb{R}^4$:

(1) $\langle Ge_0, e_1\rangle = (-1)^{|e_0|}\langle e_0, Ge_1\rangle$; (2) $\mathrm{ord}\, G \leq 1$; (3) $G^2 = 0$;
(4) $H = [Q, G] = |p|^2 \cdot 1$; (5) $\forall p \neq 0:\ \mathscr{F}_p = \mathrm{Im}\, Q(p) \oplus \mathrm{Im}\, G(p)$, $\mathrm{Ker}\, H(p) = 0$;
zero modes $(p = 0)$: scale-∞ statements (Theorem 2.4 (iii)).

Proposition 1.2 (Minimal second-order complex)**.** *On $(\mathscr{E}, Q)$ the admissible G with $[Q, G] = |p|^2$ is unique:*

$$G:\ A^* \mapsto -2A^*, \quad A \mapsto -d^*(A|_V)\ (\textit{slot } c), \quad c^* \mapsto dc^*\ (\textit{slot } A^*|_V), \quad \lambda^* \mapsto -\not{d}|_{\Sigma^*}\lambda^*,$$

$$G^2A^* = 2\,d^*(A^*|_V)\ (\textit{slot } c), \qquad G^2c^* = -2\,dc^*\ (\textit{slot } A|_V), \qquad \mathrm{rk}\, Q(p) + \mathrm{rk}\, G(p) = 27 + 28 > 54.$$

[check: `B_formulations`, `C_axioms` NEG] Conditions (3), (5) fail for every admissible choice.

2 Formulation (A): the non-minimal second-order complex

Definition 2.1 (Trivial pair, reduced pair, gauge fermion)**.**

- Trivial pair: $\bar{c}$ (degree +1, odd), b (degree 0, even), antifields $\bar{c}^*$ (degree 0), b^* (degree 1); weights $2, 2, 2, 2$; even members $\bar{c}^*$, b; $S_b = S + \int(b, \bar{c}^*)$.
- Reduced pair: $\mathscr{E}^{\bar{c}} = \{\bar{c}, \bar{c}^*\}$, $S' = S - \frac{1}{4}\int(\bar{c}^*, \bar{c}^*)$, $Q' = Q \oplus (\bar{c}^* \mapsto -\frac{1}{2}\bar{c}^*$ in slot $\bar{c})$.
- Gauge fermion $\Psi = \int(\bar{c}, d^*A|_V)$; $\Phi_\Psi = 1 + N$, $N\bar{c} = -d\bar{c}$ (slot $A^*|_V$), $NA = d^*(A|_V)$ (slot $\bar{c}^*$);

$$S_\Psi = S' \circ \Phi_\Psi = S\big(c, A, \lambda; c^*, A^* - d\bar{c}, \lambda^*\big) - \tfrac{1}{4}\int\big(\bar{c}^* + d^*A|_V, \bar{c}^* + d^*A|_V\big), \qquad I_\Psi = I \circ j_\Psi.$$

Proposition 2.2 (Canonical transformations and the elimination of b).

- *(i)* $1+N$ *with* $N\bar c = a_1 d\bar c$, $NA = a_2 d^* A|_V$ *preserves the pairing iff* $a_1 = -a_2$. *[check]*
- *(ii)* $1+N_b$, $N_b\bar c = \kappa \bar c$ *(slot* b^**),* $N_b b = \kappa' b$ *(slot* $\bar c^*$*), preserves the pairing iff* $\kappa' = -\kappa$. *At* $\kappa = -1$,

$$S_b \circ (1+N_b) = S + \int (b, \bar c^*) + \int (b,b) = S' + \int \big(b + \tfrac12 \bar c^*, b + \tfrac12 \bar c^*\big), \qquad E_b = \bar c^* + 2b = 0 \;\Rightarrow\; S'.$$

- *(iii) With* $S_b^{(1)}(v) = -\int (v.\bar c, \bar c^*) + \int (v.b, b^*)$, $S_b^{(2)}(x_1,x_2) = -\int (u_{12}.\bar c, b^*)$, *the* b*-dependence after* $1+N_b$ *at* $b^* = 0$ *is* $\int \big((v.\bar c, b) + (v.b, \bar c)\big) = \int \mathrm{div}(\cdot)$, *and the elimination of* b *gives the reduced extension of Definition 6.1 exactly. [check:* `E_equivariant_nm_sector`, $\gamma_b = -1$*]*

slot	Q (linear part of S_Ψ)	Q^{GF}
c	$-dc$ $(A\|_V)$, d^*dc $(\bar c^*)$	0
A	$-\frac12(d^*d + dd^*)A$ (A^*), $-\frac12 d^* A\|_V$ $(\bar c)$	0
λ	$\not{d}\lambda$ (λ^*)	0
$\bar c$	$-d^*d\bar c$ (c^*)	0
A^*	$d^*(A^*\|_V)$ (c^*)	$-2A^*$ (A)
λ^*	0	$-\not{d}\|_{\Sigma^*}\lambda^*$ (λ)
$\bar c^*$	$-\frac12 \bar c^*$ $(\bar c)$, $-\frac12 d\bar c^*$ $(A^*\|_V)$	$\bar c^*$ (c)
c^*	0	$-c^*$ $(\bar c)$

Theorem 2.3 (Existence and uniqueness). *Replace* $-\frac14$ *by* $\frac12\alpha$ *in* S'. *On* $\mathscr{E} \oplus \mathscr{E}^{\bar c}$ *the admissible* G *with* $[Q_\alpha, G] = |p|^2$ *and condition (1) is unique for every* $\alpha \neq 0$*; its coefficients on* $A \to c$, $A^*|_V \to \bar c^*$, $\bar c \to A|_V$, $c^* \to A^*|_V$ *are* $\alpha + \frac12$, $-\frac{2\alpha+1}{2\alpha}$, $\frac{2\alpha+1}{2\alpha}$, $\alpha + \frac12$, *and*

$$G^2 \;\propto\; \frac{(2\alpha+1)^2}{4\alpha}, \qquad G^2 = 0 \iff \alpha = -\tfrac12 \text{ (Feynman)}, \qquad G|_{\alpha=-1/2} = Q^{GF} \text{ of the table.}$$

[check: `B_formulations`*]*

Theorem 2.4 (All conditions hold).

- *(i) (1)–(4):* $(Q^{GF})^2 = 0$, $\mathrm{ord} \le 1$, *adjointness, and* $H = QQ^{GF} + Q^{GF}Q = d^*d + dd^*$ *on every component of every slot:*

$$\begin{aligned}
&Q^{GF}Qc = Q^{GF}(d^*dc \text{ in } \bar c^*) = d^*dc, \qquad Q^{GF}QA = -2\cdot(-\tfrac12)(d^*d+dd^*)A,\\
&Q^{GF}Q\bar c = -(-d^*d\bar c),\\
&QQ^{GF}A^* + Q^{GF}QA^* = (d^*d+dd^*)A^* + \underbrace{d^*A^*|_V - d^*A^*|_V}_{\text{slot } \bar c},\\
&QQ^{GF}\bar c^* + Q^{GF}Q\bar c^* = d^*d\bar c^* + \underbrace{(-d\bar c^* + d\bar c^*)}_{\text{slot } A|_V}.
\end{aligned}$$

- *(ii) (5): at* $p = (\frac32, -2, 1, \frac53)$, $\mathrm{rk}\, Q = \mathrm{rk}\, Q^{GF} = 28$, $\mathrm{rk}[Q \,|\, Q^{GF}] = 56$ *(per colour);* $\mathrm{Ker}\, Q^{GF}(p) = \mathrm{Im}\, Q^{GF}(p) =$ *span of the slots* $c, A, \lambda, \bar c$ *(Lagrangian);* $\mathrm{Ker}\, Q = \mathrm{Im}\, Q$.
- *(iii) Scale* ∞ *([C07, Lemma 6.5.1] on* $\mathbb{R}^4$*): with* $P = Q^{GF}|p|^{-2}$ *on moment-free Schwartz sections,*

$$QP + PQ = 1, \qquad P^2 = 0, \qquad PQP = P, \qquad QPQ = Q, \qquad P|_{\mathrm{Im}\, Q} = \big(Q|_{\mathrm{Im}\, Q^{GF}}\big)^{-1};$$

the zero-mode classes on compactly supported sections are those of the minimal complex $(-\int (c^*, X)$, *harmonic scalars, polynomial Weyl solutions,* $(a,x)(b,-)$ *with* $(a,b) = 0$*);* $\mathscr{E}^{\bar c}$ *contributes none,* $\bar c^* \mapsto -\frac12 \bar c^*$ *being invertible at* $p = 0$.

- *(iv) Real slice:* $\frac{1}{2}\langle A, QA\rangle = -\frac{1}{4}\int\big((dA,dA)+(d^*A|_V,d^*A|_V)\big) = -\frac{1}{4}\int(A,HA)$*, the A-block* $-\frac{1}{2}|p|^2\cdot 1_{10}$*; the former block* $-\frac{1}{2}(|p|^2 - p\otimes\iota_p)$ *on* $A|_V$ *has the longitudinal zero eigenvalue. [check]*

Proposition 2.5 (Propagator)**.** $P(\varepsilon,T)=\int_\varepsilon^T (Q^{GF}\otimes 1)K_t\,dt$*, components times* $(e^{-\varepsilon|p|^2}-e^{-T|p|^2})/|p|^2$*:*

$$A\otimes A:\ 2(-,-)^{-1}\otimes g_{V_{10}}^{-1},\qquad \lambda\otimes\lambda:\ (-,-)^{-1}\,i\rho(p)|_{\Sigma^*\to\Sigma},\qquad c\otimes\bar c:\ -(-,-)^{-1},\qquad \bar c\otimes c:\ +(-,-)^{-1};$$

all components with an antifield leg vanish; graded symmetry

$$P_{ij}(p)=(-1)^{|i||j|}P_{ji}(-p).$$

[check: `C_propagator`*]*

Proposition 2.6 (b kept)**.** *On* $\mathscr{E}\oplus\{\bar c,b,\bar c^*,b^*\}$ *with* $S_b\circ(1+N_b)\circ\Phi_\Psi$*, for every frame parameter* κ *the admissible G with* $[Q,G]=|p|^2$ *and (1) is unique and* $G^2=0$*; at* $\kappa=-1$ *it maps antifields to fields, with the extra components* $A^*\mapsto d^*A^*|_V$ *(slot b),* $b^*\mapsto db^*$ *(slot* $A|_V$*). b enters no vertex, so every graph agrees with the reduced pair. [check]*

3 Formulation (B): first order

Definition 3.1 (First-order complex)**.** Gauge sector (abelian; λ, $A|_{W_{\mathbb{R}}^\perp}$ unchanged): c (w 0), $A|_V$ (w 1), $B\in\Omega^2_+$ (w 2), B^* (w 2), $A^*|_V$ (w 3), c^* (w 4); $S_0=\int\big((B,d_+A)+(B,B)+(dc,A^*)\big)$, $D^a(A)=\eta^a_{mn}\partial_m A_n$; eliminating B: $-\frac{1}{4}\int(D,D)=-\frac{1}{4}\int(dA,dA)+\text{topological}$.

Proposition 3.2 (No scale-free square-zero operator)**.** *The admissible components are* $A\to c$*,* $B^*\to A$*,* $A^*\to A$*,* $A^*\to B$*,* $c^*\to A^*$ *(*$B^*\to B$ *needs two derivatives,* $B\to c$ *and* $c^*\to B^*$ *have no covariant form).* $[Q,G]=|p|^2$ *has the unique solution* $(h_1,\dots,h_5)=(1,-1,-2,1,1)$*, and* $G^2\neq 0$ *(*$A^*\to A\to c$*). Costello's Hermitian adjoint* $Q^\dagger$ *for a metric* $\mathrm{diag}(k_c,k_A,k_B,k_{B^*},k_{A^*},k_{c^*})$ *has* $(Q^\dagger)^2=0$ *and*

$$H|_{B\to B}=\frac{k_{A^*}|p|^2+4k_{B^*}}{k_B},\qquad H|_{B\to A}=\frac{2i\,k_{B^*}}{k_A}\,p\quad(\textit{one component}),$$

a mass term $\mu^2=4k_{B^*}/k_B$ *and first-order mixing: an infrared scale. [check:* `B_formulations` *first-order]*

Remark 3.3 ($\mathfrak{psl}(4|4)$ on B)**.** With $\tilde B=B-F_{A+}$, $S_{1\mathrm{st}}=S+\int(\tilde B,\tilde B)$ up to the topological term; the point transformation $(A,B)\mapsto(A,\tilde B)$, lifted to antifields, maps the extension $S^{(1)}+\int(x.\tilde B,\tilde B^*)$ ($\tilde B$ a weight-2 self-dual form, inert under $\Sigma\oplus\Sigma^*$) to a local solution of the equivariant master equation, with $\delta_x B=\delta_x(F_{A+})+x.(B-F_{A+})$, e.g. $\delta_\xi B=-\big(d_A\Gamma(\xi,\lambda)|_V\big)_+$. The extension is consistent; the gauge sector keeps the non-invariance of Theorem 6.6, and (B) adds the scale of Proposition 3.2.

4 Formulation (C) and the recommendation

Remark 4.1 (Other options)**.**

- Covariant gauges $\xi\neq 1$: $H|_{A_V}=|p|^2+(\xi^{-1}-1)p\otimes\iota_p$ has non-scalar symbol, condition (4) fails; enter only as the formal deformation of Proposition 6.7.
- Landau-type (b as Lagrange multiplier, no b^2): the (A_L,b) block is first order, (4) fails.
- Hermitian adjoint of the second-order Q: $H\ni(d^*d)^2$, order four, (4) fails.
- Holomorphic gauge $\bar\partial^*$: a twist as input, excluded by R2; breaks Spin(4).

- Background-field gauge $\int(\bar{c}, d^*_{\bar{A}}A)$: same structure as (A) at $\bar{A}=0$, not needed at one loop with the present observables.

	(3) $G^2=0$	(4) $H=\lvert p\rvert^2$, scale-free	(5) at $p\neq 0$	extra fields
minimal second order	no	yes	no	–
(A) reduced $(\bar{c},\bar{c}^*)$, Feynman	yes	yes	yes	$\bar{c},\bar{c}^*$
(A) with b	yes	yes	yes	$\bar{c},b,\bar{c}^*,b^*$
(B) scale-free	no	yes	no	B,B^*
(B) Hermitian adjoint	yes	no (mass μ^2)	yes	B,B^*

Proposition 4.2 (Recommendation)**.** *Formulation (A) with the reduced pair $(\bar{c},\bar{c}^*)$ in the gauge-fermion frame: (i) the unique admissible square-zero operator (Theorem 2.3); (ii) all conditions on $\mathbb{R}^4$ with no scale (Theorem 2.4); (iii) two extra slots, propagators $A\otimes A$, $\lambda\otimes\lambda$, $c\otimes\bar{c}$; (iv) arity zero is the former theory pulled back (Theorem 5.1); (v) the composite lines along Δ,P,M,R vanish and the remaining gauge-fixing variations are local vertices (Propositions 6.3, 6.4).*

5 The former construction inside (A)

Theorem 5.1 (Pullback and compression)**.** $j_\Psi=\mathrm{pr}\circ\Phi_\Psi$, $j_\Psi(c,A,\lambda,\bar{c};c^*,A^*,\lambda^*,\bar{c}^*)=(c,A,\lambda;c^*,A^*-d\bar{c},\lambda^*)$, $Q_C^{GF}=\Phi_\Psi Q^{GF}\Phi_\Psi^{-1}$. *Then*

$$
\begin{aligned}
&j_\Psi Q_\Psi = Q\, j_\Psi, \qquad j_\Psi j_\Psi^\dagger = 1 \;\Rightarrow\; (j_\Psi\otimes j_\Psi)K_t = K_t,\\
&j_\Psi Q^{GF} j_\Psi^\dagger = G_{\mathrm{former}} \;\Rightarrow\; (j_\Psi\otimes j_\Psi)P_\Psi(\varepsilon,T) = P_{\mathrm{former}}(\varepsilon,T),\\
&I_\Psi = I\circ j_\Psi, \qquad W\big(P_\Psi, J\circ j_\Psi\big) = W\big(P_{\mathrm{former}},J\big)\circ j_\Psi,\\
&\Delta_t(J\circ j_\Psi) = (\Delta_t J)\circ j_\Psi,\\
&\mathrm{pr}\, Q_C^{GF}\,\iota = G_{\mathrm{former}}, \qquad G_{\mathrm{former}}^2 = -\mathrm{pr}\, Q_C^{GF}\,\pi_{\bar{c}}\, Q_C^{GF}\,\iota,\\
&(Q_C^{GF})_{\bar{c}\to\bar{c}^*} = -2d^*d \quad (\textit{order } 2).
\end{aligned}
$$

[check: `C_frames`] The Costello frame satisfies (1), (3), (4), (5) and violates (2); the gauge-fermion frame satisfies all.

Corollary 5.2 (Arity zero)**.** $I_\Psi^{CT}[\varepsilon]=I^{CT}[\varepsilon]\circ j_\Psi$, *the defect vertex and the obstruction are pullbacks, and $F\circ j_\Psi$ solves the new master equation iff F solves the former one: $\frac{9}{4}\varepsilon^{-1}$, the log class,* Ob_2, $\frac{27}{16}$, Ob_3, *F_3 are unchanged, with $A^*|_V\to A^*|_V-d\bar{c}$ in outputs (e.g. $-\log\varepsilon\int(dc,A^*)\to-\log\varepsilon\int(dc,A^*-d\bar{c})$).*

6 Equivariant structure

Definition 6.1 (Extension to the antighost)**.** $v.\bar{c}=v(\bar{c})+\frac{1}{2}(\mathrm{div}\,v)\bar{c}$, $v.\bar{c}^*=v(\bar{c}^*)+\frac{1}{2}(\mathrm{div}\,v)\bar{c}^*$ for conformal Killing v; $\mathfrak{so}(W_{\mathbb{R}}^{\perp})$, Σ, Σ^* act trivially on $\bar{c},\bar{c}^*$; $u_{12}(x)=\Gamma(\epsilon_1(x),\epsilon_2(x))|_V$.

$$
\begin{aligned}
&S^{(1)\prime}(v) = S^{(1)}(v) - \int (v.\bar{c},\bar{c}^*), \qquad S^{(2)\prime}(x_1,x_2) = S^{(2)}(x_1,x_2) - \int \big(\bar{c},u_{12}(\bar{c})\big),\\
&S_\Psi^{(1)}(v) = S^{(1)}(v)\circ j_\Psi - \int \big(v.\bar{c},\bar{c}^* + d^*A|_V\big), \qquad S_\Psi^{(1)}(x) = S^{(1)}(x)\circ j_\Psi \;(x\in\mathfrak{so}(W)\oplus\Sigma\oplus\Sigma^*),\\
&S_\Psi^{(2)}(x_1,x_2) = S^{(2)}(x_1,x_2)\circ j_\Psi - \int \big(\bar{c},u_{12}(\bar{c})\big).
\end{aligned}
$$

Theorem 6.2 (Equivariant master equation). $d_{CE}S_\bullet + \frac{1}{2}\{S_\bullet, S_\bullet\} = 0$ *for* $S_\bullet = S' + S^{(1)\prime} + S^{(2)\prime}$ *and* $S_\bullet = S_\Psi + S^{(1)}_\Psi + S^{(2)}_\Psi$, *every* $\mathfrak{g}$. *Proof: disjoint fields* $\Rightarrow S_\bullet = S_{\mathfrak{psl}(4|4)} + S^{\bar c}$ *with zero cross bracket; Theorem c2-master for the first; for* $S^{\bar c}$*:*

$$\{-\tfrac{1}{4}(\bar c^*)^2, -(v.\bar c, \bar c^*)\} = \tfrac{1}{2}\int (v.\bar c^*, \bar c^*) = \tfrac{1}{4}\int \operatorname{div}\big(v(\bar c^*, \bar c^*)\big) = 0, \qquad [v., w.] = [v, w].,$$
$$\{-\tfrac{1}{4}(\bar c^*)^2, -(\bar c, u(\bar c))\} = \tfrac{1}{2}\int (\bar c^*, u(\bar c)) - \tfrac{1}{2}\int (\bar c, u(\bar c^*)) = \int (u.\bar c, \bar c^*) = -S^{(1)\prime}(u)|_{\bar c},$$
$$\{-(v.\bar c, \bar c^*), -(\bar c, u(\bar c))\} = \int (\bar c, [v, u](\bar c)), \quad u_{[v,x_2],x_3} + u_{x_2,[v,x_3]} = [v, u_{23}], \qquad \{S^{(2)\prime}|_{\bar c}, S^{(2)\prime}|_{\bar c}\} = 0;$$

Φ_Ψ *preserves the pairing and commutes with* d_{CE}. *The coefficient* -1 *is unique* ($+1$ *fails,* 0 *fails). [check:* `E_equivariant_abelian` *26 components* $\times$ *frames min/C/*Ψ*, NEG;* `E_equivariant_nm_sector`*;* $\mathfrak{su}(2)$ $\{S_\Psi, S_\Psi\} = 0$ *in* `D_engine`*]*

Proposition 6.3 (The variations of the gauge fermion). $s_x\Psi :=$ *antifield-free part of* $S^{(1)}_\Psi(x)$*:*

$$s_v\Psi = -\int \big((v.A|_V, d\bar c) + (v.\bar c, d^*A|_V)\big) = -\tfrac{1}{2}\int \big(\bar c, \iota_{\operatorname{grad}\operatorname{div} v}A|_V\big),$$
$$s_v\Psi = \begin{cases} 0 & P_b,\ M_\omega,\ \Delta \\ 4\int(\bar c, \iota_b A|_V) & K_b \end{cases}, \qquad s_R\Psi = 0,$$
$$s_\epsilon\Psi = \int \big(\Gamma(\epsilon,\lambda)|_V, d\bar c\big),$$
$$(s_\epsilon\Psi).:\ \lambda \mapsto -d^*\Gamma(\epsilon,\lambda)|_V\ (\bar c^*), \quad \bar c \mapsto \rho(d\bar c)\epsilon\ (\lambda^*),$$
$$(s_{K_b}\Psi).:\ \bar c \mapsto -4(b,-)\bar c\ (A^*|_V), \quad A \mapsto 4\iota_b A\ (\bar c^*),$$
$$S^{(2)}_\Psi \supset -\int \big(\bar c, \iota_{u_{12}} d\bar c\big).$$

[check: `F_generators`*]*

Proposition 6.4 (Generators on the gauge fixing). $x.$ *the operator of the quadratic part of* $S^{(1)}_\Psi(x)$ *(*$\{F, \ell\} = \ell \circ F.$*);* $G_{\varepsilon T} = \int_\varepsilon^T e^{-tH}dt$*;* $[x., Q] = 0$ *for all* $x.$

$(P\star)^2 = (Q^{GF})^2 G^2_{\varepsilon T} = 0, \qquad P \star x.P\star = P \star [x., Q^{GF}]G_{\varepsilon T} = P \star (s_x\Psi).P\star,$
$[x., Q^{GF}] = 0 \ \ (P_b, M_\omega, R_r), \qquad [\Delta., Q^{GF}] = -4Q^{GF},\ [\Delta., H] = -4H,$
$[v., Q^{GF}] = -\frac{1}{2}(\operatorname{div} v)Q^{GF} + \frac{1}{4}\rho(d\operatorname{div} v)|_{\lambda^*\to\lambda} + [(s_v\Psi)., Q^{GF}],$
$[(s_{K_b}\Psi)., Q^{GF}]:\ A \mapsto -4\iota_b A\ (c), \quad \bar c \mapsto -8(b,-)\bar c\ (A|_V), \quad A^*|_V \mapsto -8\iota_b A^*\ (\bar c^*), \quad c^* \mapsto 4(b,-)c^*\ (A^*|_V),$
$[\epsilon., Q^{GF}] = [\epsilon., Q^{GF}]_{A^*\to\lambda,\ \lambda^*\to A}$
$\quad + [(s_\epsilon\Psi)., Q^{GF}]_{\lambda\to c,\ \bar c\to\lambda,\ \lambda^*\to\bar c^*,\ c^*\to\lambda^*};$
$P \star x.P\star = 0 \ \ (P_b, M_\omega, \Delta, R_r),$
$P \star x.P\star \neq 0 \ \ (\xi, \eta:\ \lambda$*–c line;* $K_b:\ A|_V$*–c line through* $4\iota_b$*).*

[check: `F_generators`*]*

Proposition 6.5 (Two descriptions). *Costello frame: vertices* $S', S^{(1)\prime}, S^{(2)\prime}$ *(no* $s_x\Psi$*), propagator* $P_C = (\Phi_\Psi \otimes \Phi_\Psi)P_\Psi$*, line* $P_C \star [x.', Q^{GF}_C]G_{\varepsilon T}$*; gauge-fermion frame: vertices* $\circ\Phi_\Psi$ *(with* $s_x\Psi$*,* $-\int(\bar c, u_{12}\bar c)$*), propagator* P_Ψ*:*

$$W(P_C, J) \circ \Phi_\Psi = W(P_\Psi, J \circ \Phi_\Psi), \qquad P_C \star [x.', Q^{GF}_C]G_{\varepsilon T} = \Phi_\Psi\, P_\Psi \star (s_x\Psi).P_\Psi \star \Phi^{-1}_\Psi.$$

Theorem 6.6 (No invariant gauge fixing along Σ, Σ^*, K). *For $\Psi' = \int(\bar c, \mathcal{G})$, $\mathcal{G}$ local, invariant, weight 2, linear part fixing the gauge: $\mathcal{G}_1 = \kappa d^*A|_V$, $\kappa \neq 0$, and for every admissible action on $\bar c$*

$$s_\epsilon \Psi'\big|_{\bar c\,\cdot\, field} = -\kappa \int (\bar c, d^*\Gamma(\epsilon,\lambda)|_V) \neq 0, \qquad s_K \Psi'\big|_{\bar c\,\cdot\, field} = -\kappa \int \big(\bar c, \tfrac{1}{2}\iota_{\operatorname{grad}\operatorname{div} v} A|_V + \tfrac{w-2}{4}(\operatorname{div} v) d^*A|_V\big) \neq 0.$$

Proposition 6.7 (Gauge parameter: Nielsen identity). $S_\Psi \to S_\Psi + uG_\alpha$, $G_\alpha = \frac{1}{2}\int(\bar c^* + d^*A|_V)^2$*:*

$$G_\alpha = -\big(Q + \{I_\Psi, -\}\big) \int \big(\bar c^* + d^*A|_V, \bar c\big) \qquad [check],$$

the image of $\{\int(\bar c^*, \bar c), -\frac{1}{4}\int(\bar c^*)^2\} = -\frac{1}{2}\int(\bar c^*)^2$. With uG_α a formal quadratic vertex (Q^{GF}, H of $u = 0$), ∂_u of a family of one-loop quantisations is s-closed with exact classical part; spectra of $\Lambda^{(1)}(\Delta)$ on cohomology and the classes $[Q_1^\xi]$ are u-independent (Theorems c8-invariants (iii), c8-twist). The same holds for $\Psi \to \Psi + u\Psi_1$: $\partial_u S_\Psi = \{S_\Psi, \Psi_1\}$. In finite form, in units: $\frac{27}{16}\int(d^*A|_V)^2 \equiv -\frac{27}{16}\int\big((\bar c^*, \bar c^*) + 2(\bar c^*, d^*A|_V)\big)$ modulo s-exact.

7 Two-component forms (Weinberg–Srednicki)

Components of Definition c8-twocomp; $\epsilon_\alpha{}^i = \xi_\alpha{}^i - i x_m \sigma^m_{\alpha\dot\alpha}\bar\eta^{\dot\alpha i}$, $\bar\epsilon_{\dot\alpha i} = \bar\xi_{\dot\alpha i} + i x_m \eta^\alpha{}_i \sigma^m_{\alpha\dot\alpha}$, $\Gamma(\epsilon_1,\epsilon_2)|^m_V = i(\bar\epsilon_{2\dot\alpha i}\bar\sigma^{m\dot\alpha\alpha}\epsilon_{1\alpha}{}^i + \bar\epsilon_{1\dot\alpha i}\bar\sigma^{m\dot\alpha\alpha}\epsilon_{2\alpha}{}^i)$ [check].

$$S_\Psi \supset \int d^4x \Big\{ -\tfrac{1}{8}(F_{mn}, F_{mn}) - \tfrac{1}{4}(\partial_m A_m, \partial_n A_n) - (D_m c, \partial_m \bar c) + (D_m c, A^*_m) - \tfrac{1}{4}(\bar c^*, \bar c^*) + \tfrac{1}{2}(\bar c^*, \partial_m A_m)\Big\},$$

$$\tfrac{1}{\hbar} S_\Psi \supset \int d^4x \Big\{ -\tfrac{1}{4} F^a_{mn} F^a_{mn} - \tfrac{1}{2}(\partial_m A^a_m)^2 - 2(D_m c)^a \partial_m \bar c^a \Big\},$$

$$\mathfrak{su}(N): \ A = -igA^aT^a, \ c = -igc^aT^a, \ \bar c = -ig\bar c^a T^a,$$

$$(Q^{GF}e)_{A_m} = -2A^*_m, \quad (Q^{GF}e)_{A_{ij}} = -2A^*_{ij}, \quad (Q^{GF}e)_c = \bar c^*, \quad (Q^{GF}e)_{\bar c} = -c^*,$$

$$(Q^{GF}e)_{\lambda_\alpha{}^i} = i\sigma^m_{\alpha\dot\alpha}\partial_m \bar\lambda^{*\dot\alpha i},$$

$$\langle A_m A_n\rangle : \ 2(-,-)^{-1}\delta_{mn}, \quad \langle c\,\bar c\rangle : \ -(-,-)^{-1}, \quad \langle \bar c\, c\rangle : \ (-,-)^{-1},$$

$$\langle \lambda_\alpha{}^i \bar\lambda_{\dot\beta j}\rangle : \ -(-,-)^{-1} p_m \sigma^m_{\alpha\dot\beta}\delta^i_j \quad \Big(\times \tfrac{e^{-\varepsilon|p|^2}}{|p|^2}\Big),$$

$$s_\epsilon \Psi = i\int d^4x \big(\bar\lambda_{\dot\alpha i}\bar\sigma^{m\dot\alpha\alpha}\epsilon_\alpha{}^i + \bar\epsilon_{\dot\alpha i}\bar\sigma^{m\dot\alpha\alpha}\lambda_\alpha{}^i, \ \partial_m \bar c\big),$$

$$s_{K_b}\Psi = 4\int d^4x\, (\bar c, b_m A_m),$$

$$S^{(2)}_\Psi(x_1,x_2) \supset -\int d^4x\, \big(\bar c, \Gamma(\epsilon_1,\epsilon_2)|^m_V \partial_m \bar c\big),$$

Weinberg: $\not{d} = \gamma^\mu_W \partial_\mu = -i\gamma^\mu_S \partial_\mu$ on the Dirac column.

8 Migration plan (detail in migration_plan.json, 84 items)

status	results

pullback, identical	Ch4: I^{CT} ($\frac{9}{4}\varepsilon^{-1}\int(A,A)$, $\log\varepsilon[-(dA,dA)+4(\lambda,\not{d}\lambda)-(dc,A^*)]$), log class $\log\varepsilon\, s(\mathcal{N}\circ j_\Psi)$, coupling 0, $\mathrm{Ob}_2=-\frac{27}{8}\int(d^*dc,d^*A|_V)$, $F_2=\frac{27}{16}\int(d^*A|_V)^2$ and its Q-closed directions, Prop c4-K. Ch5: $O(\varepsilon)$ of letters and scalar bilinears. Ch6: Callan–Symanzik form, field redefinition, L_ε graphs. Ch9 §1: $\mathrm{Ob}_3=(\frac{11}{4}+\frac{19}{4}\log\frac{3}{2})(G_1+G_3)+\frac{27}{8}G_2-9\log\frac{3}{2}G_5$, F_3 family (β_k,κ_k), Wess–Zumino. Ch10: all pair kernels of the counterterm map, Thm c10-brst. Arity one along Σ, Σ^*, $\mathfrak{so}(W)$: defect cycles, $\mathrm{Ob}^{(1)}(\xi),\mathrm{Ob}^{(1)}(\eta)$, their solutions; two-body parts of Q_1,S_1 ($-2\log\frac{3}{2}\mathcal{B}_1$, $4\log\frac{3}{2}(\mathcal{B}_2+\mathcal{B}_3)$). Twisted theory along square-zero ξ: $[Q_1^\xi]$, chiral bilinears, BK comparison.
rewritten	Ch3 (this replacement); Def c7-vdot, Prop c7-cov, c7-dil, c7-susy (with $\bar{c}$); Prop c8-twocomp; Lemma c9-SDelta; Prop c9-conj ($e^{\hbar\partial_P}S^{(1)}(\Delta)=(S^{(1)}(\Delta)+\hbar P\star v.)e^{\hbar\partial_P}$); Lemma c11-lines; placements in Ch11, Ch12.
disappears	composite kernels $2P\star v.P\star\propto(Q^{GF})^2$ along Δ, P_b, $\mathfrak{so}(V)$ (Def/Prop c9-composite, Rem c9-Pi, Prop c9-triangle, Ch11 §§1–3 composite pair kernels and their locality), θ_P, the finite part ψ' for $P_1=0$ (Prop c9-scheme), Rem c7-internal.
recompute (R10)	arity one along Δ, P_b, $\mathfrak{so}(V)$, K_b (Thm c9-quadratic: composite part $[\frac{1}{3}(dA,dA)-\frac{1}{3}(\lambda,\not{d}\lambda)-\frac{2}{3}(dc,A^*)]$ gone, special-edge part 4×log coefficient kept, new $\bar{c}$ placements); θ and all t_x (Thm c9-theta2, c12-fix, c12-ar2, Cor c12-inv); D_1, P_1, M_1, K_1 representatives (Thm c9-D1, c10-D1, c12-F0, c12-D1); arity two along pairs with $u_{12}\neq0$ and along (Δ,P_b), $(K_b,P_{b'})$, (v,ϵ); $\log T$ along K_b (Prop c8-Tsafe, Thm c8-K: former coefficient $\frac{3}{8\pi^2}$); reflection Θ_n on $\bar{c},\bar{c}^*$. Expected from the structure: $D_1=4\times$(map of Thm c10-brst) with the new sector coefficient fixed by the new (Δ,P_b) equation.
invariant, must agree	Konishi $\gamma_K=\frac{3g^2N}{4\pi^2}$ ($D_1=-48$ units on the trace), $20'$: 0; families $D_1=-8$ units $\times E_H$ ($\sum_l H_{l,l+1}$ of [B03c]); dilatation coefficients 0, -24; twist classes; one-loop algebra relations on cohomology.

9 Machine checks

group	content (all exact; NEG = negative control, fails as required)
A	Clifford relations of ρ on V_{10}, symmetry for $(-,-)$, chirality, 't Hooft identity, two-component $\Gamma\|_V$.
B	minimal: unique G, $G^2 \neq 0$; first order: unique scale-free G, $G^2 \neq 0$, Hermitian adjoint with mass term; reduced: $a_1 = -a_2$, α-family, $G^2 \propto (2\alpha+1)^2/(4\alpha)$; b kept: unique G for every κ, $G^2 = 0$.
C	$Q^2 = 0$, $(Q^{GF})^2 = 0$, (1), (2), (4) on all 56 components per colour; ranks $(28, 28, 56)$; NEG former ranks $(27, 28, 54)$; Lemma 6.5.1 identities; projectors; real slice; NEG former longitudinal zero mode; propagator and graded symmetry; K_0 represents 1; frames: $Q_\Psi = \Phi_\Psi^{-1} Q' \Phi_\Psi$, j_Ψ chain map, $j_\Psi j_\Psi^\dagger = 1$, compression, G^2_{former} identity, order-two entry of Q_C^{GF}, adjointness and ranks of Q_C^{GF}.
D	jet engine: linear part of $\{S_0, -\}$ equals Q on every coordinate (frames min, C, Ψ); $\mathfrak{su}(2)$ $\{S_\Psi, S_\Psi\} = 0$; three NEG.
E	equivariant master equation, abelian, 26 components (arity one: $P, M, \Delta, K, K', R, \xi, \xi', \eta, \eta'$; arity two: even–even, even–odd, odd–odd incl. R): min, C, Ψ, inert $\bar{c}$ pass; NEG $\gamma_2 = 0$ and inert $\bar{c}$ with $\gamma_2 = -1$ fail exactly on the four odd–odd components; antighost sector arities 1–4; trivial pair $(\bar{c}, b)$, $\gamma_b = -1$, elimination of b; $G_\alpha = -s(\cdot)$.
F	$H = -\partial^2$ on all slots (position space); $[x., Q^{GF}]$ for P, M, R, Δ, K; $Q^{GF} x . Q^{GF} = 0$ (P, M, Δ, R), $\neq 0$ (K, ξ, η), $= Q^{GF}(s_x\Psi).Q^{GF}$ (all); closed forms of $s_x\Psi$ and of $[v., Q^{GF}]$.